Introduction to Operating Systems

3. Processes

2026-10-14

USO Team

Bibliography

1. **Brian Ward**, *How LINUX Works*, 3rd Edition, No Starch Press, 2021
    - Chapter 1 - *The Big Picture*
        - Section 1.3.1 - *Process Management*
    - Chapter 2 - *Basic Commands and Directory Hierarchy*
        - Section 2.14 - *Shell Input and Output*
        - Section 2.16 - *Listing and Manipulating Processes*
        - Section 2.17.2 - *Working with Symbolic Links*
    - Chapter 3 - *Devices*
        - Section 3.1 - *Device Files*
    - Chapter 4 - *Disks and Filesystems*
        - Section 4.2.12 - *Special-Purpose Filesystems*
    - Chapter 6 - *How User Space Starts*
        - Sections 6.1 - 6.6
    - Chapter 8 - *A Closer Look at Processes and Resource Utilization*
        - Sections 8.1 - 8.3
2. **Razvan Deaconescu, Razvan Rughinis, Mihai Carabas, Alexandru Radovici**, *Utilizarea Sistemelor de Operare*, Printech 2021, download
    - Chapter 2 - *Utilizarea sistemului de fișiere*
        - Section 2.3.4.1 - *Crearea legăturilor simbolice*
        - Section 2.4 - *Redirectarea intrării sau ieșirii*
    - Chapter 3 - *Pachete software*
        - Sections 3.1 - 3.3
    - Chapter 4 - *Procese*
        - Sections 4.1 - 4.5, 4.8 - 4.9
    - Chapter 8 - *Componente hardware*
        - Section 8.6 - *Abstractizarea dispozitivelor în Linux*
    - Chapter 9 - *Pornirea sistemului*
        - Section 9.6 - *Pornirea init și a serviciilor de startup în Linux*

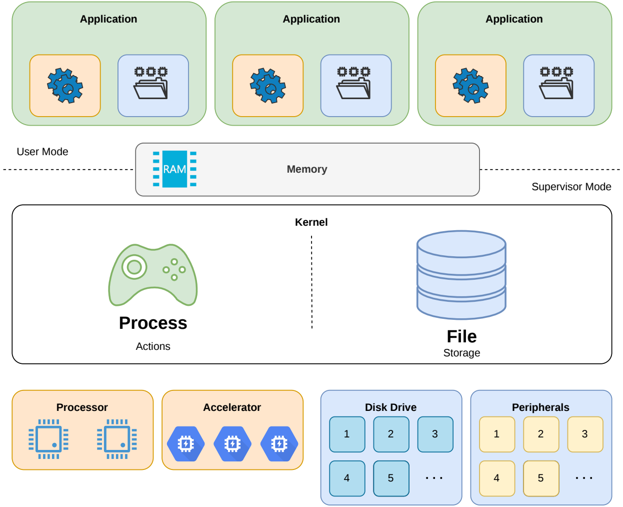
Operating System: abstractions

Actions

- *Applications*
- use the *Processor* and *Accelerators* (GPU, Neural Engine, etc)

Data

- everything is a file
- peripherals are viewed as files (*POSIX*)
  - `/dev/input/keyboard` - keyboard
  - `/dev/fb` - screen (framebuffer)
  - `/dev/sda` - Disk Drive A (the first drive)


Application
Application
Application
User Mode
RAM
Memory
Supervisor Mode
Kernel
Process
Actions
File
Storage
Processor
Accelerator
Disk Drive
1
2
3
4
5
...
Peripherals
1
2
3
4
5
...

Installing Applications

App repositories

Bibliography

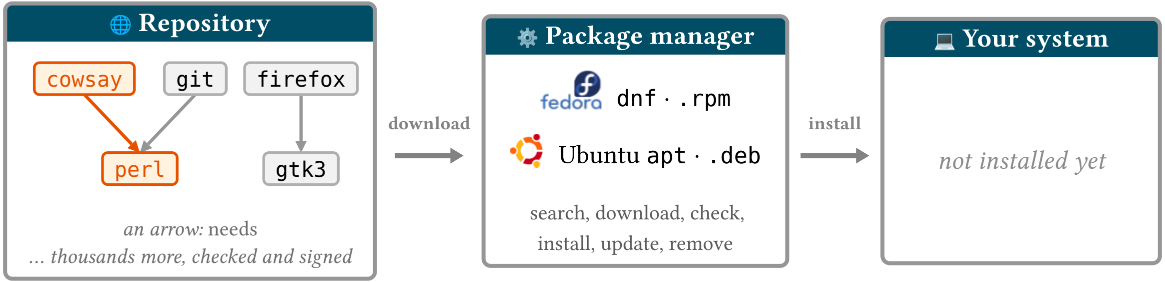
For this section

1. **Razvan Deaconescu, Razvan Rughinis, Mihai Carabas, Alexandru Radovici**, *Utilizarea Sistemelor de Operare*, Printech 2021, download
    - Chapter 3 - *Pachete software*
        - Section 3.1 - *Pachete software*
        - Section 3.2 - *Gestiunea pachetelor*
        - Section 3.3 - *Gestiunea pachetelor în Linux*

App Repositories

The Linux app store, long before the app stores

- applications come as **packages**: the program, its files and the list of packages it needs (*dependencies*)

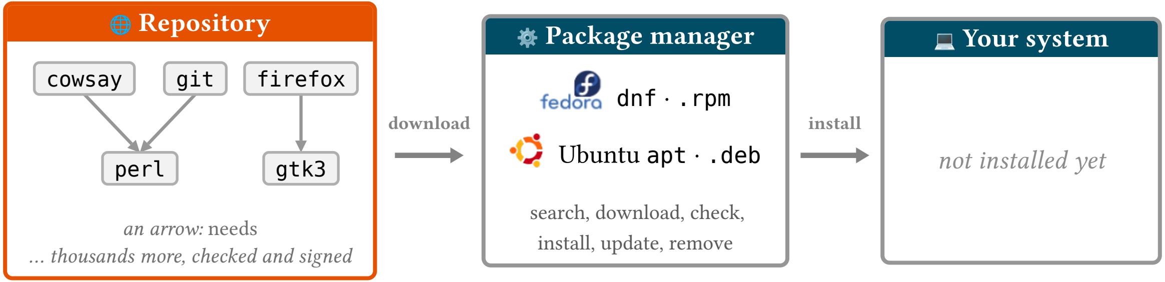


🌐 Repository
cowsay
git
firefox
perl
gtk3
an arrow: needs
… thousands more, checked and signed
download
⚙ Package manager
fedora dnf · .rpm
Ubuntu apt · .deb
search, download, check, install, update, remove
install
💻 Your system
not installed yet

App Repositories

The Linux app store, long before the app stores

- applications come as **packages**: the program, its files and the list of packages it needs (*dependencies*)
- the distribution keeps them in a **repository**: a server with thousands of checked and signed packages



🌐 Repository

`cowsay` → `perl`
`git` → `perl`
`firefox` → `gtk3`

an arrow: needs
… thousands more, checked and signed

download

⚙ Package manager

fedora `dnf` · `.rpm`
Ubuntu `apt` · `.deb`

search, download, check,
install, update, remove

install

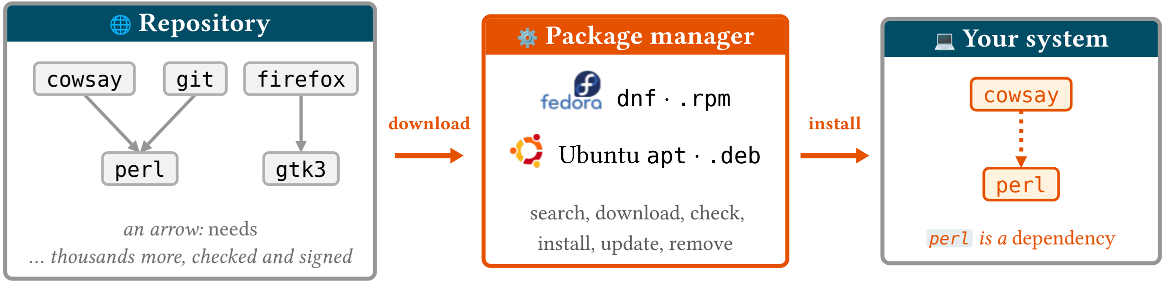
💻 Your system

not installed yet

App Repositories

The Linux app store, long before the app stores

- applications come as **packages**: the program, its files and the list of packages it needs (*dependencies*)
- the distribution keeps them in a **repository**: a server with thousands of checked and signed packages
- the **package manager** searches, downloads, installs, updates and removes them, dependencies included



Repository

`cowsay` `git` `firefox`

`perl` `gtk3`

an arrow: needs

… thousands more, checked and signed

download

Package manager

dnf · .rpm

Ubuntu apt · .deb

search, download, check, install, update, remove

install

Your system

`cowsay`

`perl`

`perl` *is a* dependency

Fedora: `dnf`

Search and install `cowsay`

```
$ dnf search cowsay
Matched fields: name (exact)
 cowsay.noarch     Configurable speaking/thinking cow
Matched fields: name
 xcowsay.x86_64    Displays a cute cow and message on your desktop
```

💡 `dnf search` looks for the name in the repositories: `cowsay` is in the `fedora` repository

Fedora: `dnf`

Search and install `cowsay`

```
$ dnf search cowsay
Matched fields: name (exact)
 cowsay.noarch     Configurable speaking/thinking cow
Matched fields: name
 xcowsay.x86_64    Displays a cute cow and message on your desktop
$ sudo dnf install cowsay
Package Arch   Version        Repository      Size
Installing:
 cowsay noarch 0:3.8.4-5.fc44 fedora      80.5 KiB
Is this ok [y/N]: y
```

💡 installing changes the system for all the users: it needs `sudo` (the *administrator*), and asks before it starts

Ubuntu: `apt`

Search and install `cowsay`

```
$ apt search cowsay
cowsay/noble 3.03+dfsg2-8 all
  configurable talking cow
xcowsay/noble 1.6-1build2 amd64
  Graphical configurable talking cow
```

💡 the same idea, another package manager: `apt search` looks in the Ubuntu repositories (`noble` is Ubuntu 24.04)

Ubuntu: `apt`

Search and install `cowsay`

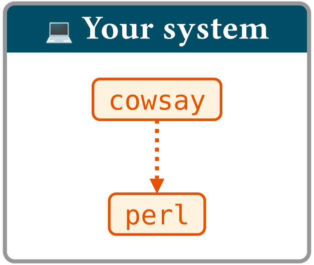
```
$ apt search cowsay
cowsay/noble 3.03+dfsg2-8 all
  configurable talking cow
xcowsay/noble 1.6-1build2 amd64
  Graphical configurable talking cow
$ sudo apt install cowsay
The following NEW packages will be installed:
  cowsay libgdbm-compat4t64 libgdbm6t64 libperl5.38t64
  libtext-charwidth-perl netbase perl perl-modules-5.38
Do you want to continue? [Y/n] y
```

💡 `cowsay` is written in Perl: `apt` installs the **dependencies** too, `perl` and its libraries

Dependencies Are Packages Too

After installing `cowsay`, `perl` is on the system

```
$ which cowsay perl
/usr/games/cowsay
/usr/bin/perl
```

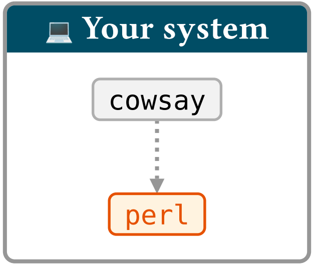


💡 `apt` installed both: `cowsay`, and `perl` because `cowsay` needs it

Dependencies Are Packages Too

After installing `cowsay`, `perl` is on the system

```
$ which cowsay perl
/usr/games/cowsay
/usr/bin/perl
$ perl -e 'print "hello from perl\n"'
hello from perl
```

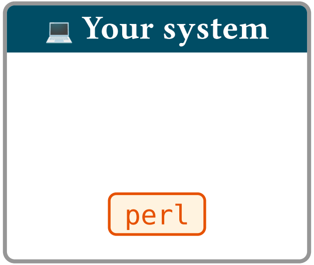


💡 `perl` is a normal package: you, or any other program, can use it **by itself**

Dependencies Are Packages Too

After installing `cowsay`, `perl` is on the system

```
$ which cowsay perl
/usr/games/cowsay
/usr/bin/perl
$ perl -e 'print "hello from perl\n"'
hello from perl
$ sudo apt remove cowsay
The following packages will be REMOVED:
  cowsay
```



💻 **Your system**

`perl`

💡 removing `cowsay` does not remove `perl`; `sudo apt autoremove` removes the dependencies that nothing needs anymore

Package Managers Cheat Sheet

Fedora and Ubuntu

	fedora	**Ubuntu**
search	`dnf search NAME`	`apt search NAME`
details	`dnf info NAME`	`apt show NAME`
install	`sudo dnf install NAME`	`sudo apt install NAME`
remove	`sudo dnf remove NAME`	`sudo apt remove NAME`
remove unused dependencies	`sudo dnf autoremove`	`sudo apt autoremove`
update everything	`sudo dnf upgrade`	`sudo apt update` `sudo apt upgrade`

💡 `apt update` only refreshes the list of packages, `apt upgrade` installs the new versions; `dnf` refreshes the list by itself

Process

From a file to an action

Bibliography

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1. **Brian Ward**, *How LINUX Works*, 3rd Edition, No Starch Press, 2021
   - Chapter 1 - *The Big Picture*
     - Section 1.3.1 - *Process Management*
   - Chapter 6 - *How User Space Starts*
     - Sections 6.2 - 6.3 - *Identifying Your init, systemd*
   - Chapter 8 - *A Closer Look at Processes and Resource Utilization*
     - Section 8.1 - *Tracking Processes*
2. **Razvan Deaconescu, Razvan Rughinis, Mihai Carabas, Alexandru Radovici**, *Utilizarea Sistemelor de Operare*, Printech 2021, download
   - Chapter 4 - *Procese*
     - Sections 4.1 - 4.3 - *Programe și procese, Resursele și atributele unui proces, Ierarhia de procese*
     - Section 4.9 - *Anexa: Internele pornirii unui proces. Loading*
   - Chapter 9 - *Pornirea sistemului*
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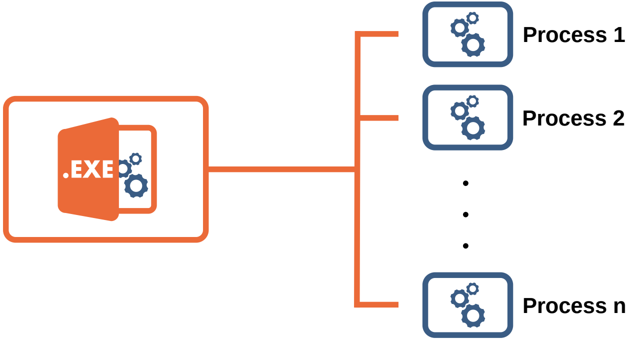
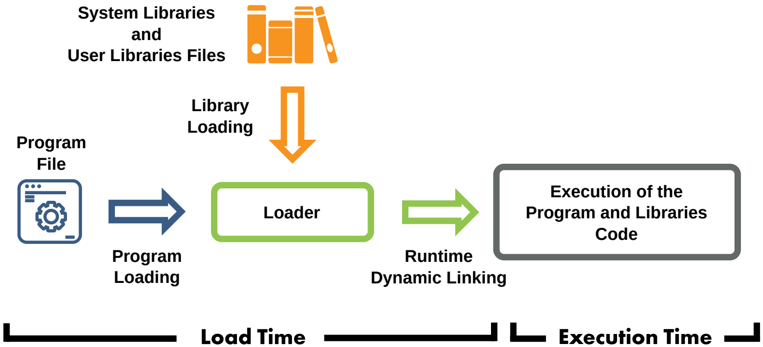
Executable vs Process

Files to *actions*

📁 one executable file ➡️ multiple processes ⚙️⚙️…⚙️


.EXE
Process 1
Process 2
Process n

Loading an Executable



System Libraries
and
User Libraries Files

Library
Loading

Program
File

Program
Loading

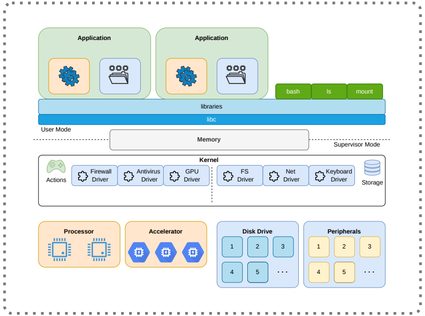
Loader

Runtime
Dynamic Linking

Execution of the
Program and Libraries
Code

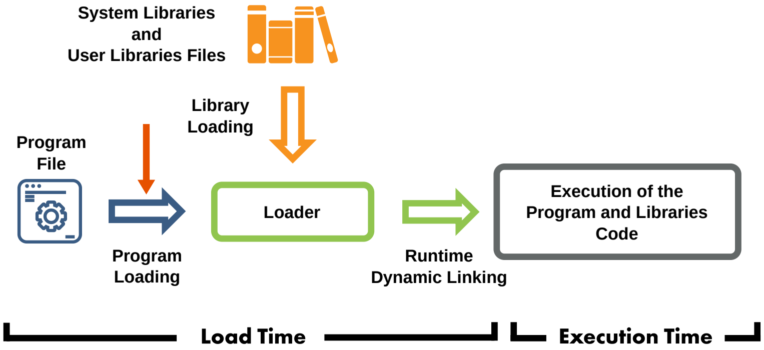
Load Time

Execution Time


Application
Application
bash
ls
mount
libraries
libc
User Mode
Memory
Supervisor Mode
Kernel
Actions
Firewall Driver
Antivirus Driver
GPU Driver
FS Driver
Net Driver
Keyboard Driver
Storage
Processor
Accelerator
Disk Drive
1
2
3
4
5
...
Peripherals
1
2
3
4
5
...

- 💡 the **loader** turns the program file into a running process, in a few steps

Loading an Executable



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Loading

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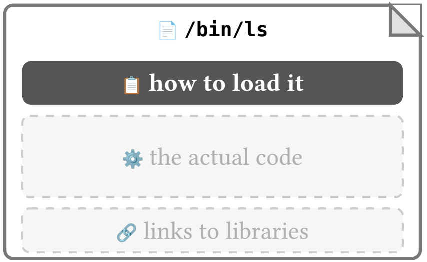


📄 `/bin/ls`

📋 **how to load it**

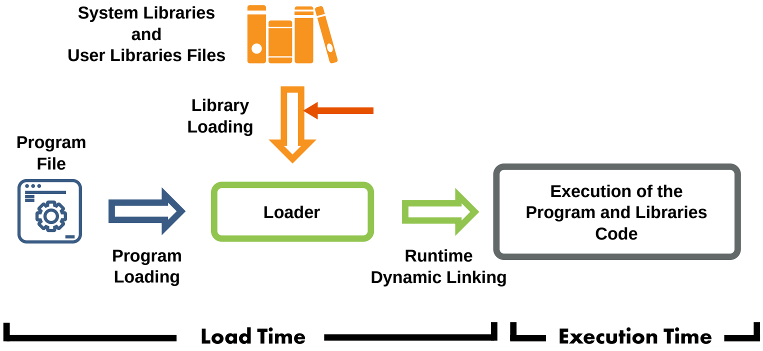
⚙ the actual code

🔗 links to libraries

Executable File

💡 **program loading**: the loader reads the executable and copies its code and data into memory

Loading an Executable



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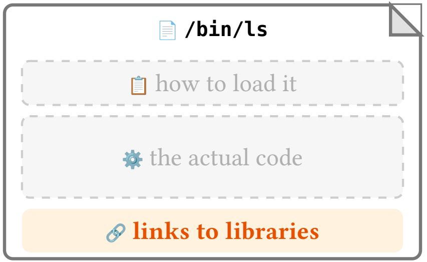


📄 `/bin/ls`

📋 how to load it

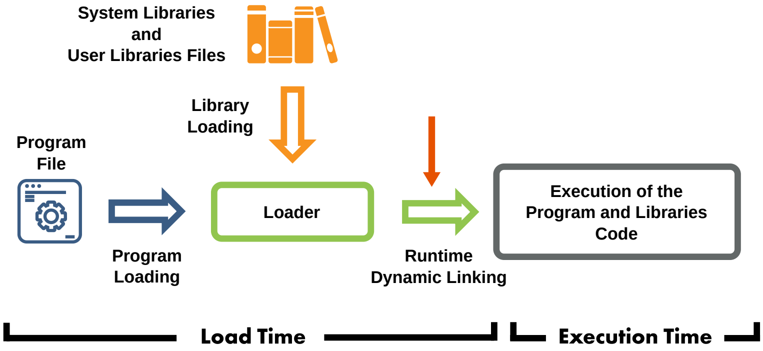
⚙ the actual code

🔗 **links to libraries**

Executable File

💡 **library loading**: it finds the libraries the program needs (`libc.so.6`, ...) and loads them too

Loading an Executable



System Libraries
and
User Libraries Files

Library
Loading

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Dynamic Linking

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Load Time

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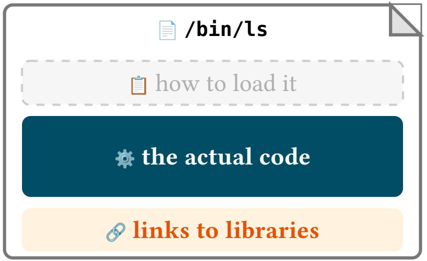


📄 `/bin/ls`

📋 how to load it

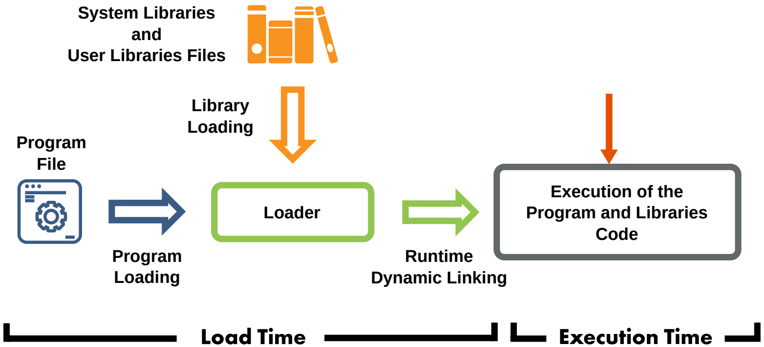
⚙ **the actual code**

🔗 **links to libraries**

Executable File

💡 **runtime dynamic linking**: it connects the program’s calls to the functions inside the libraries

Loading an Executable



System Libraries and User Libraries Files

Library Loading

Program File

Program Loading

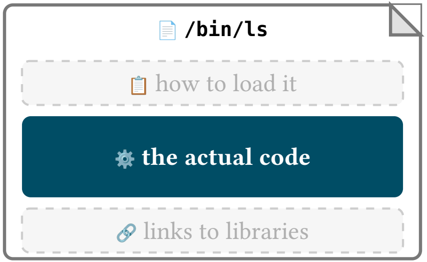
Loader

Runtime Dynamic Linking

Execution of the Program and Libraries Code

Load Time

Execution Time



📄 `/bin/ls`

📋 how to load it

⚙ **the actual code**

🔗 links to libraries

Executable File

💡 **execution**: the processor jumps to the start of the program, the code of the program and of the libraries runs

Process’ Resources

Identification

- *PID*: **P**rocess **ID**
- *PPID*: **P**arent **P**rocess **ID**

⚙️ Actions

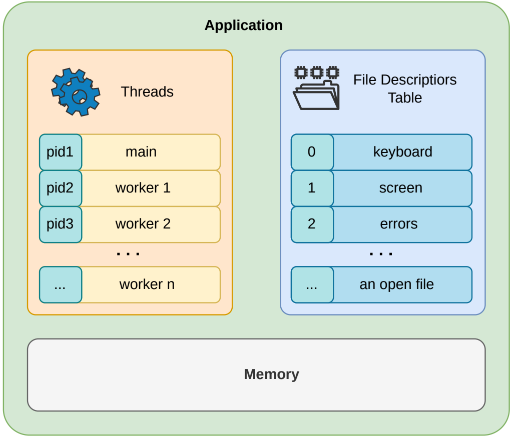
- one or multiple threads executing in parallel

💾 Data

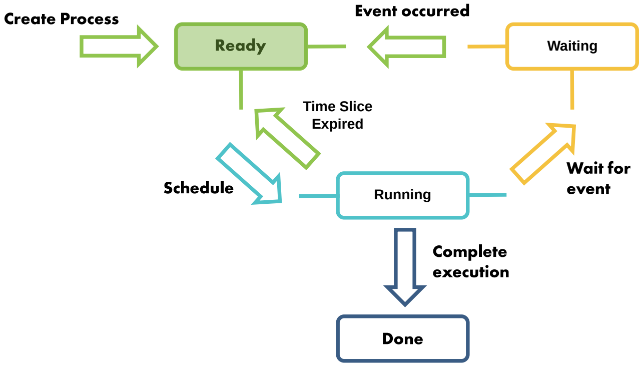
- access to files
  - actual files
  - peripheral files (`/dev/...`)

🧠 Memory

- code
- variables


Application
Threads
pid1
main
pid2
worker 1
pid3
worker 2
...
...
worker n
File Descriptors Table
0
keyboard
1
screen
2
errors
...
...
an open file
Memory

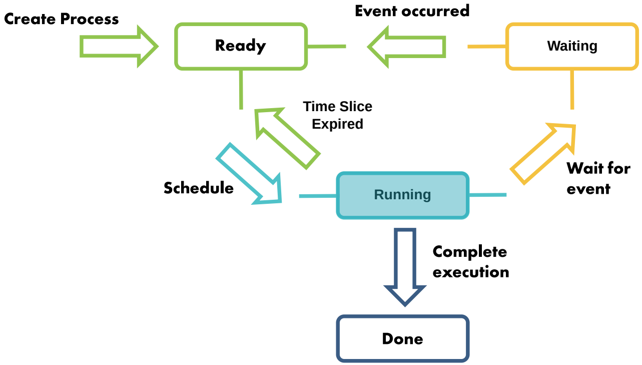
Process States


Create Process
Ready
Event occurred
Waiting
Time Slice Expired
Schedule
Running
Wait for event
Complete execution
Done

```
$ ps -e -o pid,stat,comm
    PID STAT COMMAND
      1 Ss   systemd
    ...
   4100 Ss   bash
   4210 S    sleep
   4211 R    yes
   4213 Sl   myapp
   4214 Z    myapp-worker <defunct>
   4220 R+   ps
```

Ready - has everything it needs, waits for a free **processor**
in `ps` it is `R`, like **Running**: `ps` cannot tell them apart

Process States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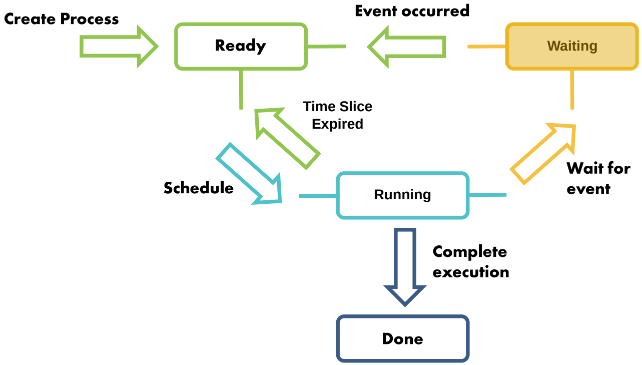
   4210 S    sleep
   4211 R    yes
   4213 Sl   myapp
   4214 Z    myapp-worker <defunct>
   4220 R+   ps
```

Running - its code runs on a processor; when its *time slice* expires, it goes back to **Ready**

`R` in `ps`: `yes` computes all the time, `ps` runs while it makes the list

Process States

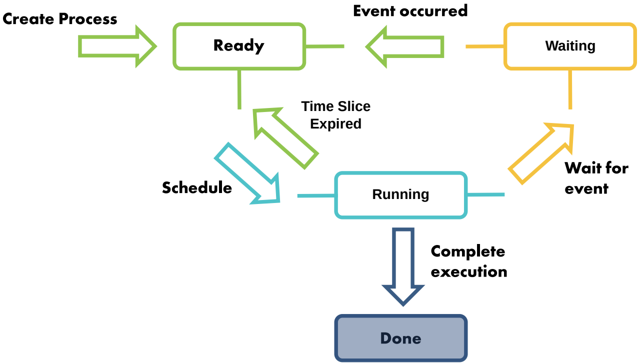

Create Process
Ready
Event occurred
Waiting
Time Slice Expired
Schedule
Running
Wait for event
Complete execution
Done

```
$ ps -e -o pid,stat,comm
    PID STAT COMMAND
      1 Ss   systemd
    ...
   4100 Ss   bash
   4210 S    sleep
   4211 R    yes
   4213 Sl   myapp
   4214 Z    myapp-worker <defunct>
   4220 R+   ps
```

Waiting - waits for an **event**: data from the disk, a key press, a network packet; it does not use the processor

`S` (*sleeping*) in `ps`: `bash` waits for you to type, `sleep` waits for its timer

Process States


Create Process
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Wait for event
Complete execution
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```
$ ps -e -o pid,stat,comm
    PID STAT COMMAND
      1 Ss   systemd
    ...
   4100 Ss   bash
   4210 S    sleep
   4211 R    yes
   4213 Sl   myapp
   4214 Z    myapp-worker <defunct>
   4220 R+   ps
```

Done - it has finished (`exit`), it waits for the parent to read its *exit code*

`Z` (*zombie*[1]) in `ps`: `myapp-worker` has finished, but its parent, `myapp`, has not read its exit code yet

---

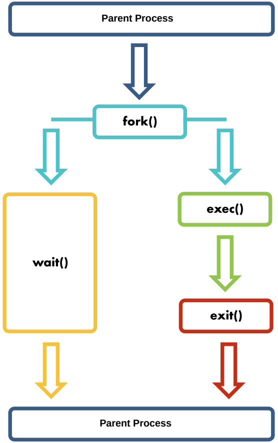
[1] *zombie* processes are explained later, in *Zombie Process*

Creating a Process

Forking a parent and loading an executable

Child

Parent


Parent Process
fork()
wait()
exec()
exit()
Parent Process

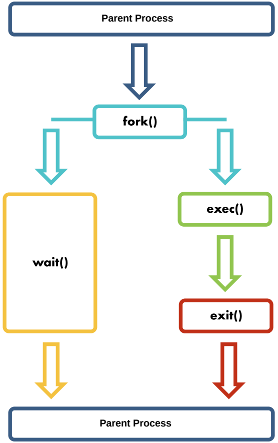
Creating a Process

Forking a parent and loading an executable

The parent process calls `fork` - *makes a **clone** of the parent*

Child

Parent


Parent Process
fork()
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Parent Process

Creating a Process

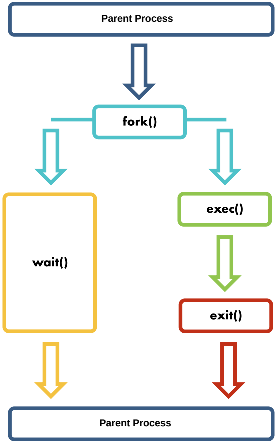
Forking a parent and loading an executable

The parent process calls `fork` - *makes a **clone** of the parent*

from now on we have **two identical processes**

Child

Parent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Creating a Process

Forking a parent and loading an executable

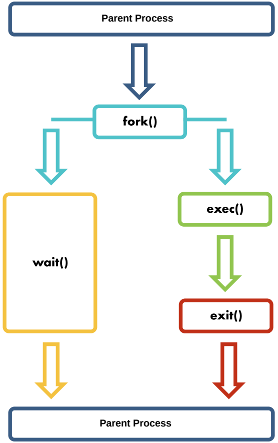
The parent process calls `fork` - *makes a **clone** of the parent*

from now on we have **two identical processes**

Child

1. Calls `exec` - *replaces the executable*

Parent


Parent Process
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Parent Process

Creating a Process

Forking a parent and loading an executable

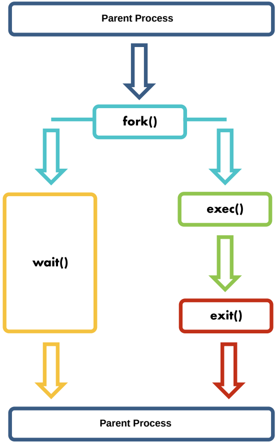
The parent process calls `fork` - *makes a **clone** of the parent*

from now on we have **two identical processes**

Child

1. Calls `exec` - *replaces the executable*
2. Eventually calls `exit` (or `return` in `main`) and returns an *exit code*

Parent


Parent Process
fork()
wait()
exec()
exit()
Parent Process

Creating a Process

Forking a parent and loading an executable

The parent process calls `fork` - *makes a **clone** of the parent*

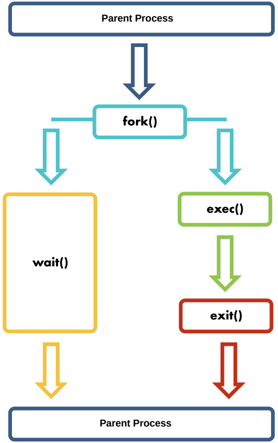
from now on we have **two identical processes**

Child

1. Calls `exec` - *replaces the executable*
2. Eventually calls `exit` (or `return` in `main`) and returns an *exit code*

Parent

1. Does other work or *may wait* for the child process


Parent Process
fork()
wait()
exec()
exit()
Parent Process

Creating a Process

Forking a parent and loading an executable

The parent process calls `fork` - *makes a **clone** of the parent*

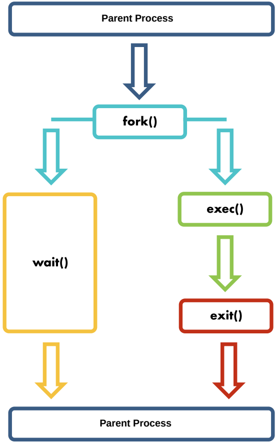
from now on we have **two identical processes**

Child

1. Calls `exec` - *replaces the executable*
2. Eventually calls `exit` (or `return` in `main`) and returns an *exit code*

Parent

1. Does other work or *may wait* for the child process
2. Eventually calls `wait` to read the *exit code* of the child



How Linux Starts

The first process

```
[kernel]  interrupt handler / scheduler
│    # The kernel’s first process (part of the kernel itself)
│    # Manages CPU scheduling and idling
```

How Linux Starts

The first process

```
[kernel]  interrupt handler / scheduler
│    # The kernel’s first process (part of the kernel itself)
│    # Manages CPU scheduling and idling
│
└── [PID 1]  systemd (called init)
      # The first user-space process started by the kernel
      # Responsible for initializing the rest of the system
      # Mounts filesystems, starts targets, and manages services
```

How Linux Starts

The first process

```
[kernel]  interrupt handler / scheduler
│   # The kernel’s first process (part of the kernel itself)
│   # Manages CPU scheduling and idling
│
├── [PID 1]  systemd (called init)
└── [PID ...]  kernel threads (kthreadd, kworker, ksoftirqd, etc.)
      # Internal kernel helpers for scheduling, I/O, and interrupts
      # These run in the background as part of the kernel itself
```

How Linux Starts

The first process

```
[kernel]  interrupt handler / scheduler
│   # The kernel’s first process (part of the kernel itself)
│   # Manages CPU scheduling and idling
│
├── [PID 1]  systemd (called init)
│      ├── [PID 3+]  systemd-udevd
│      │       # Handles dynamic device detection (e.g., USB, disks)
│
└── [PID ...]  kernel threads (kthreadd, kworker, ksoftirqd, etc.)
```

How Linux Starts

The first process

```
[kernel]  interrupt handler / scheduler
│   # The kernel's first process (part of the kernel itself)
│   # Manages CPU scheduling and idling
│
├── [PID 1]  systemd (called init)
│     ├── [PID 3+]  systemd-udevd
│     ├── [PID 5+]  NetworkManager (or systemd-networkd)
│     │       # Configures and manages network interfaces
│
└── [PID ...]  kernel threads (kthreadd, kworker, ksoftirqd, etc.)
```

How Linux Starts

The first process

```
[kernel]  interrupt handler / scheduler
│   # The kernel’s first process (part of the kernel itself)
│   # Manages CPU scheduling and idling
│
├── [PID 1]  systemd (called init)
│     ├── [PID 3+]  systemd-udevd
│     ├── [PID 5+]  NetworkManager (or systemd-networkd)
│     ├── [PID 8+]  gdm / sddm / lightdm (optional)
│     │       # Display manager - starts graphical login session
│
└── [PID ...]  kernel threads (kthreadd, kworker, ksoftirqd, etc.)
```

How Linux Starts

The first process

```
[kernel]  interrupt handler / scheduler
│   # The kernel’s first process (part of the kernel itself)
│   # Manages CPU scheduling and idling
│
├── [PID 1]  systemd (called init)
│     ├── [PID 3+]  systemd-udevd
│     ├── [PID 5+]  NetworkManager (or systemd-networkd)
│     ├── [PID 8+]  gdm / sddm / lightdm (optional)
│     └── [PID 10+]  user@1000.service
│             # Per-user systemd instance
│             # Manages user-level services after login
└── [PID ...]  kernel threads (kthreadd, kworker, ksoftirqd, etc.)
```

How Linux Starts

The first process

```
[kernel]  interrupt handler / scheduler
│   # The kernel's first process (part of the kernel itself)
│   # Manages CPU scheduling and idling
│
├── [PID 1]  systemd (called init)
│     ├── [PID 3+]  systemd-udevd
│     ├── [PID 5+]  NetworkManager (or systemd-networkd)
│     ├── [PID 8+]  gdm / sddm / lightdm (optional)
│     └── [PID 10+]  user@1000.service
│             ├── [PID 11+]  bash / zsh / fish
│             │       # User's shell process after login
└── [PID ...]  kernel threads (kthreadd, kworker, ksoftirqd, etc.)
```

How Linux Starts

The first process

```
[kernel]  interrupt handler / scheduler
│   # The kernel’s first process (part of the kernel itself)
│   # Manages CPU scheduling and idling
│
├── [PID 1]  systemd (called init)
│     ├── [PID 3+]  systemd-udevd
│     ├── [PID 5+]  NetworkManager (or systemd-networkd)
│     ├── [PID 8+]  gdm / sddm / lightdm (optional)
│     └── [PID 10+]  user@1000.service
│             ├── [PID 11+]  bash / zsh / fish
│             ├── [PID 12+]  Xorg / Wayland
│             │       # Graphical display server (for GUI sessions)
└── [PID ...]  kernel threads (kthreadd, kworker, ksoftirqd, etc.)
```

How Linux Starts

The first process

```
[kernel]  interrupt handler / scheduler
│   # The kernel’s first process (part of the kernel itself)
│   # Manages CPU scheduling and idling
│
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│     ├── [PID 5+]  NetworkManager (or systemd-networkd)
│     ├── [PID 8+]  gdm / sddm / lightdm (optional)
│     └── [PID 10+]  user@1000.service
│             ├── [PID 11+]  bash / zsh / fish
│             ├── [PID 12+]  Xorg / Wayland
│             └── [PID 13+]  gnome-shell / plasma / sway
│                     # User’s desktop environment or window manager
└── [PID ...]  kernel threads (kthreadd, kworker, ksoftirqd, etc.)
```

How Linux Starts

The first process

```
[kernel]  interrupt handler / scheduler
│   # The kernel's first process (part of the kernel itself)
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│
├── [PID 1]  systemd (called init)
│     ├── [PID 3+]  systemd-udevd
│     ├── [PID 5+]  NetworkManager (or systemd-networkd)
│     ├── [PID 8+]  gdm / sddm / lightdm (optional)
│     └── [PID 10+]  user@1000.service
│             ├── [PID 11+]  bash / zsh / fish
│             ├── [PID 12+]  Xorg / Wayland
│             └── [PID 13+]  gnome-shell / plasma / sway
│                     # User's desktop environment or window manager
└── [PID ...]  kernel threads (kthreadd, kworker, ksoftirqd, etc.)
```

⚠️ **`init` cannot stop!**

or the whole system will panic!

😕 Orphan Process

What happens when the parent process ends

- child processes remain without a parent
- orphan processes are *reparented* to `init` (PID 1)

```
[PID 1]  systemd (init)
│     # PID 1: init, adopts orphaned processes
│     │
│     ├── [PID 10+]  user@1000.service
│     │     │
│     │     ├── [PID 100]  myapp
│     │     │     # Application master process
│     │     │     │
│     │     │     ├── [PID 200]  myapp-worker
│     │     │     │     # Worker doing background jobs
│     │     │     └── [PID 201]  myapp-helper
│     │     │           # Helper process
...
```

😕 Orphan Process

What happens when the parent process ends

- child processes remain without a parent
- orphan processes are *reparented* to `init` (PID 1)

```
[PID 1]  systemd (init)
│     # PID 1: init, adopts orphaned processes
│     │
│     ├── [PID 10+]  user@1000.service
│     ...
├── [PID 200]  myapp-worker    <-- reparented
│     # Was child of PID 100; now adopted by PID 1
├── [PID 201]  myapp-helper    <-- reparented
│     # Also adopted by PID 1
...
```

💡 `myapp` (PID 100) has ended

😕 Orphan Process

What happens when the parent process ends

- child processes remain without a parent
- orphan processes are *reparented* to `init` (PID 1)

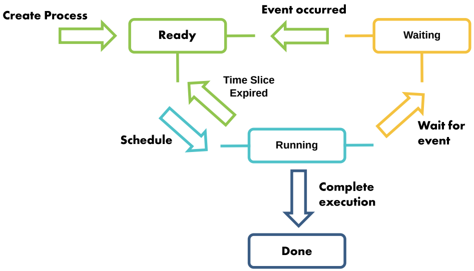
```
[PID 1]  systemd (init)
│     # PID 1: init, adopts orphaned processes
│     │
│     ├── [PID 10+]  user@1000.service
│     ...
├── [PID 200]  myapp-worker    <-- reparented
│     # Was child of PID 100; now adopted by PID 1
├── [PID 201]  myapp-helper    <-- reparented
│     # Also adopted by PID 1
...
```

⚠️ `init` has to be present to be able to reparent processes

or the whole system will panic!

🧟 Zombie Process

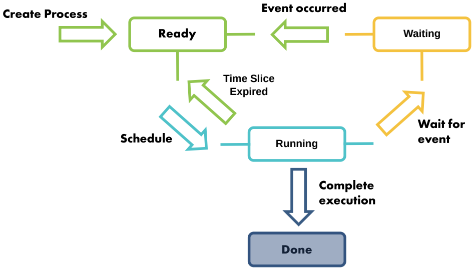
And this is why `init` cannot stop


Create Process
Ready
Event occurred
Waiting
Time Slice Expired
Schedule
Running
Wait for event
Complete execution
Done

🧟 Zombie Process

And this is why `init` cannot stop

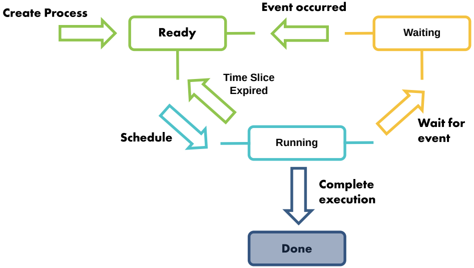
- the process will stay in `Done` and will **wait for the parent** to *read its return code* (`wait`)


Create Process
Ready
Event occurred
Waiting
Time Slice Expired
Schedule
Running
Wait for event
Complete execution
Done

🧟 Zombie Process

And this is why `init` cannot stop

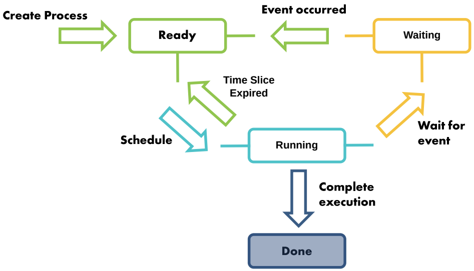
- the process will stay in `Done` and will **wait for the parent** to *read its return code* (`wait`)
- the system will **keep all the process’ resources allocated** while in `Done`


Create Process
Ready
Event occurred
Waiting
Time Slice Expired
Schedule
Running
Wait for event
Complete execution
Done

🧟 Zombie Process

And this is why `init` cannot stop

- the process will stay in `Done` and will **wait for the parent** to *read its return code* (`wait`)
- the system will **keep all the process' resources allocated** while in `Done`


Create Process
Ready
Event occurred
Waiting
Time Slice Expired
Schedule
Running
Wait for event
Complete execution
Done

- until the parent reads the return code of the child, **resources** stay **allocated**, **but not used**: 🧟 zombie

😕 Orphan and 🧟 Zombie ⁉️

Uses up resources

⚠️ `init` has to be **present** to be able to reparent processes

or the whole system will be full of 🧟 zombies

🐧 + 🧟 = 🛑

Linux does not like ***permanent*** *zombies, so it just panics when* `init` *stops.*

File Descriptors

How processes access data

Bibliography

For this section

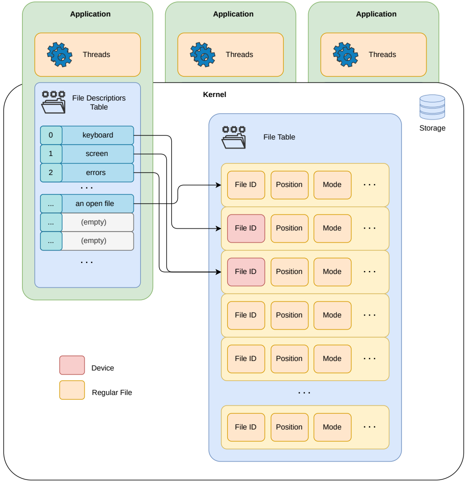
1. **Brian Ward**, *How LINUX Works*, 3rd Edition, No Starch Press, 2021
    - Chapter 8 - *A Closer Look at Processes and Resource Utilization*
        - Section 8.2 - *Finding Open Files with lsof*
2. **Razvan Deaconescu, Razvan Rughinis, Mihai Carabas, Alexandru Radovici**, *Utilizarea Sistemelor de Operare*, Printech 2021, download
    - Chapter 4 - *Procese*
        - Section 4.4.1 - *Descriptori de fișiere*
        - Section 4.8 - *Anexa: Sistemul de fișiere procfs*

File Descriptor Table

How processes access data

- each process has a *File Descriptor Table*
- open files show up in the table
- the **index** in the table is the **file descriptor**

Special Descriptors


Application
Application
Application
Threads
Threads
Threads
Kernel
Storage
File Descriptors Table
0 keyboard
1 screen
2 errors
...
... an open file
... (empty)
... (empty)
...
File Table
File ID Position Mode ...
File ID Position Mode ...
File ID Position Mode ...
File ID Position Mode ...
File ID Position Mode ...
...
File ID Position Mode ...
Device
Regular File

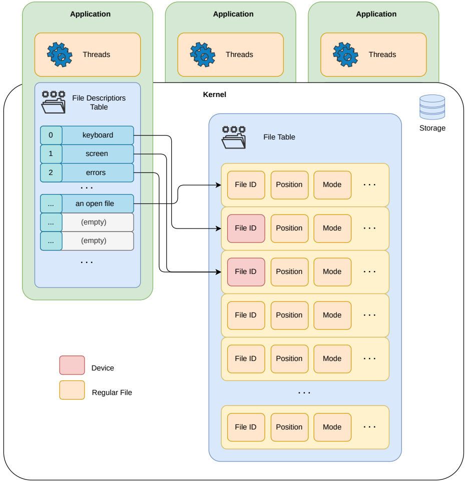
File Descriptor Table

How processes access data

- each process has a *File Descriptor Table*
- open files show up in the table
- the **index** in the table is the **file descriptor**

Special Descriptors

- `0` - *keyboard* - `stdin`


Application
Application
Application
Threads
Threads
Threads
Kernel
Storage
File Descriptors Table
0 keyboard
1 screen
2 errors
...
... an open file
... (empty)
... (empty)
...
File Table
File ID
Position
Mode
...
Device
Regular File

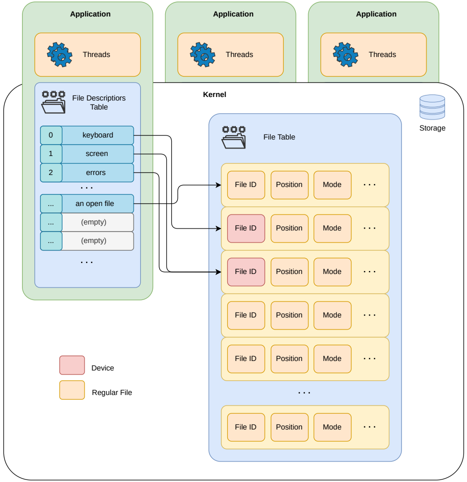
File Descriptor Table

How processes access data

- each process has a *File Descriptor Table*
- open files show up in the table
- the **index** in the table is the **file descriptor**

Special Descriptors

- `0` - *keyboard* - `stdin`
- `1` - *display* - `stdout`


Application
Threads
Application
Threads
Application
Threads
File Descriptors Table
0 keyboard
1 screen
2 errors
...
... an open file
... (empty)
... (empty)
...
Kernel
Storage
File Table
File ID
Position
Mode
...
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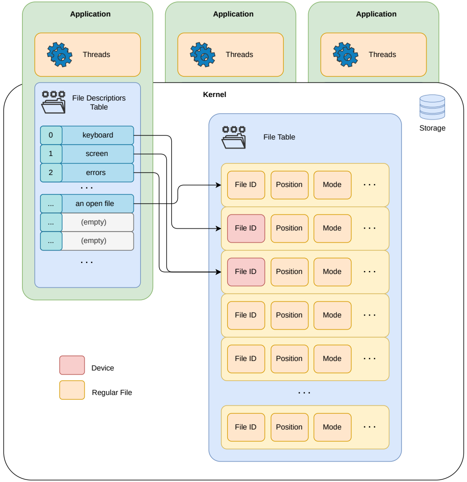
File Descriptor Table

How processes access data

- each process has a *File Descriptor Table*
- open files show up in the table
- the **index** in the table is the **file descriptor**

Special Descriptors

- `0` - *keyboard* - `stdin`
- `1` - *display* - `stdout`
- `2` - *error display* - `stderr`

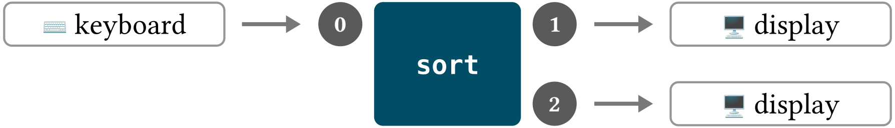

Application
Threads
Application
Threads
Application
Threads
File Descriptors Table
0
keyboard
1
screen
2
errors
...
an open file
...
(empty)
...
(empty)
Kernel
Storage
File Table
File ID
Position
Mode
Device
Regular File

`sort` reads a file

The process opens the file by itself

```
$ cat fruits.txt
cherry
apple
banana
$ sort fruits.txt    # the file name is an argument
```

FD Table of `sort`	
0	⌨️ keyboard
1	🖥️ display
2	🖥️ display



⌨️ keyboard → 0 **sort**

1 → 🖥️ display

2 → 🖥️ display

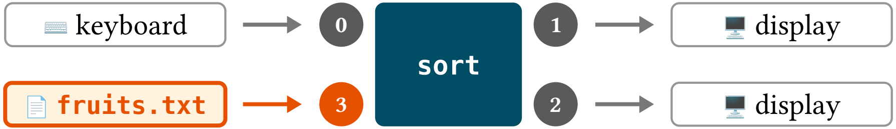
💡 the **shell** starts `sort` with `0`, `1` and `2`, and gives it `fruits.txt` as an *argument*

`sort` reads a file

The process opens the file by itself

```
$ cat fruits.txt
cherry
apple
banana
$ sort fruits.txt    # the file name is an argument
```

FD Table of `sort`	
0	⌨️ keyboard
1	🖥️ display
2	🖥️ display
3	📄 `fruits.txt`


⌨️ keyboard
0
📄 fruits.txt
3
sort
1
🖥️ display
2
🖥️ display

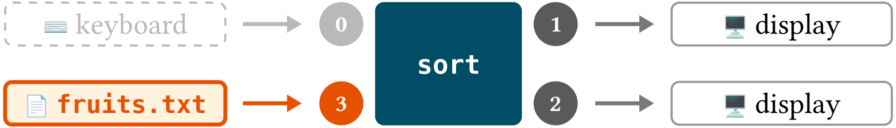
💡 `sort` calls `open("fruits.txt")`, the kernel places it at the first free index: `3`

`sort` reads a file

The process opens the file by itself

```
$ cat fruits.txt
cherry
apple
banana
$ sort fruits.txt     # the file name is an argument
apple
banana
cherry
```

FD Table of `sort`	
0	⌨️ keyboard
1	🖥️ display
2	🖥️ display
3	📄 **`fruits.txt`**



⌨️ keyboard → 0

📄 `fruits.txt` → 3

`sort`

1 → 🖥️ display

2 → 🖥️ display

💡 `sort` reads the lines from `3`, sorts them and writes them to `1`, the keyboard (`0`) is not used

Redirects

Change where a process reads and writes

Bibliography

For this section

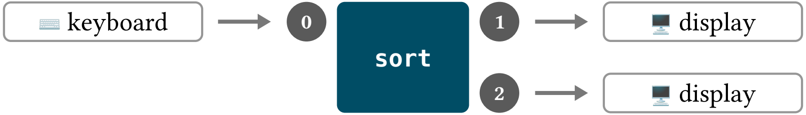
1. **Brian Ward**, *How LINUX Works*, 3rd Edition, No Starch Press, 2021
    - Chapter 2 - *Basic Commands and Directory Hierarchy*
        - Section 2.14 - *Shell Input and Output*
    - Chapter 3 - *Devices*
        - Section 3.1 - *Device Files*
2. **Razvan Deaconescu, Razvan Rughinis, Mihai Carabas, Alexandru Radovici**, *Utilizarea Sistemelor de Operare*, Printech 2021, download
    - Chapter 2 - *Utilizarea sistemului de fișiere*
        - Section 2.4 - *Redirectarea intrării sau ieșirii*
    - Chapter 4 - *Procese*
        - Section 4.4.2 - *Redirectarea în/din fișiere*
        - Sections 4.5.3 - 4.5.4 - *Înlănțuirea comenzilor, Comunicarea prin pipe-uri*
    - Chapter 8 - *Componente hardware*
        - Section 8.6 - *Abstractizarea dispozitivelor în Linux*

`<` - redirect `stdin`

The file **replaces** the keyboard as `stdin`

```
$ sort                  # reads the keyboard, until Ctrl+D
```

FD Table of `sort`	
0	⌨️ keyboard
1	🖥️ display
2	🖥️ display


⌨️ keyboard
0
sort
1
2
🖥️ display
🖥️ display

💡 `sort` reads its lines from `0` (`stdin`), which is the keyboard

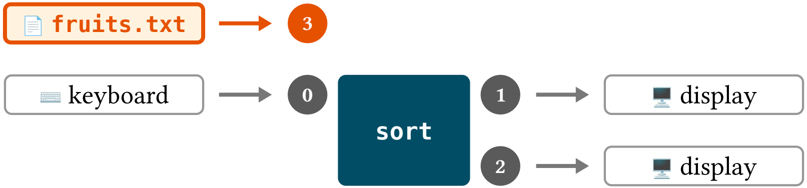
`<` - redirect `stdin`

The file **replaces** the keyboard as `stdin`

```
$ sort                # reads the keyboard, until Ctrl+D
$ sort < fruits.txt   # reads fruits.txt
```

FD Table of `sort`

0	⌨️ keyboard
1	🖥️ display
2	🖥️ display
3	📄 `fruits.txt`


📄 fruits.txt
3
⌨️ keyboard
0
sort
1
🖥️ display
2
🖥️ display

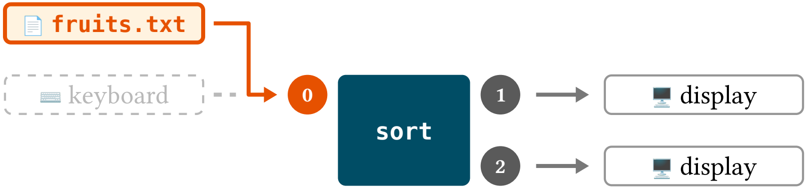
💡 the **shell** opens `fruits.txt` for `sort`, it gets the first free index: `3`

`<` - redirect `stdin`

The file **replaces** the keyboard as `stdin`

```
$ sort                 # reads the keyboard, until Ctrl+D
$ sort < fruits.txt    # reads fruits.txt
apple
banana
cherry
```

FD Table of `sort`	
0	📄 `fruits.txt`
1	🖥️ display
2	🖥️ display


📄 fruits.txt
⌨️ keyboard
0
sort
1
2
🖥️ display
🖥️ display

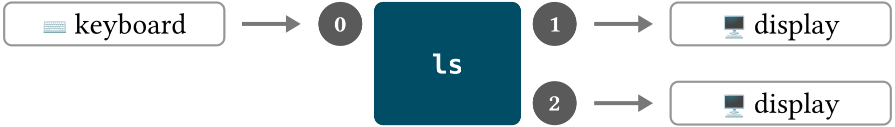
- 💡 `<` **replaces** `stdin`: `0` is connected to `fruits.txt` instead of the keyboard, `3` is not needed anymore
- 💡 `sort` does not know that it reads a file, it still just reads from `0` (`stdin`)

`>` - redirect `stdout`

The process writes into a file instead of the display

```
$ ls                          # writes on the display
Movies   notes.txt   todo.txt
```

FD Table of `ls`	
0	⌨ keyboard
1	🖥 display
2	🖥 display



⌨ keyboard → 0 `ls`

`ls` 1 → 🖥 display

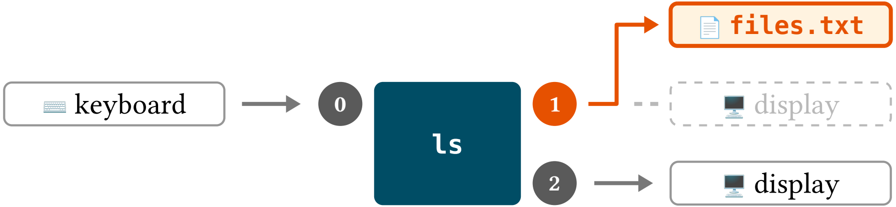
`ls` 2 → 🖥 display

`>` - redirect `stdout`

The process writes into a file instead of the display

```
$ ls                    # writes on the display
Movies   notes.txt   todo.txt
$ ls > files.txt        # nothing shows up
```

FD Table of `ls`	
0	⌨️ keyboard
1	📄 `files.txt`
2	🖥️ display


⌨️ keyboard
0
ls
1
2
📄 files.txt
🖥️ display
🖥️ display

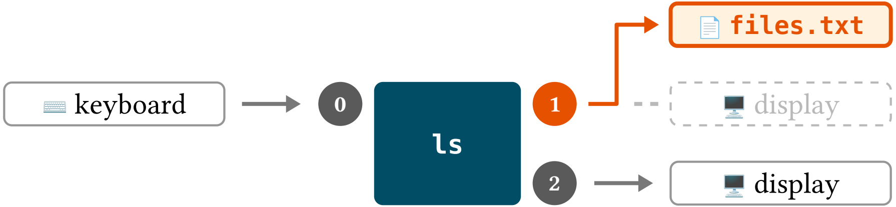
⚠️ `>` *empties* `files.txt` first (or creates it), use `>>` to **append** to it

`>` - redirect `stdout`

The process writes into a file instead of the display

```
$ ls                        # writes on the display
Movies   notes.txt   todo.txt
$ ls > files.txt            # nothing shows up
$ ls /nope > files.txt
ls: cannot access '/nope': No such file or directory
```

FD Table of `ls`	
0	⌨️ keyboard
1	📄 **`files.txt`**
2	🖥️ display


⌨️ keyboard
0
ls
1
📄 files.txt
🖥️ display
2
🖥️ display

💡 only `1` (`stdout`) is redirected, errors still go to the display through `2`, use `2>` for them

`>>` - append to `stdout`

```
$ echo one > log.txt    # empty the file, then write
```

📄 log.txt

```
one
```

	the shell opens the file with	writes go
`>`	`open("log.txt", O_WRONLY | O_CREAT | O_TRUNC)`	after the file is **emptied**
`>>`	`open("log.txt", O_WRONLY | O_CREAT | O_APPEND)`	at the **end**, always

💡 `>` uses `O_TRUNC`: the file is emptied before `echo` writes `one`

`>>` - append to `stdout`

```
$ echo one > log.txt     # empty the file, then write
$ echo two >> log.txt    # write at the end
```

📄 `log.txt`

```
one
two
```

	the shell opens the file with	**writes go**
`>`	`open("log.txt", O_WRONLY | O_CREAT | O_TRUNC)`	after the file is **emptied**
`>>`	`open("log.txt", O_WRONLY | O_CREAT | O_APPEND)`	at the **end**, always

💡 `>>` uses `O_APPEND` instead: nothing is deleted, the kernel moves every write to the end of the file

`>>` - append to `stdout`

```
$ echo one > log.txt     # empty the file, then write
$ echo two >> log.txt    # write at the end
$ cat log.txt
one
two
```

📄 `log.txt`

```
one
two
```

	the shell opens the file with	**writes go**
`>`	`open("log.txt", O_WRONLY | O_CREAT | O_TRUNC)`	after the file is **emptied**
`>>`	`open("log.txt", O_WRONLY | O_CREAT | O_APPEND)`	at the **end**, always

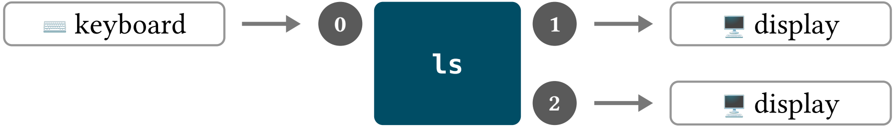
💡 useful for logs: every command adds its lines, nothing is lost

`2>` - redirect `stderr`

The errors go into a file instead of the display

```
$ ls /nope                # the error is on the display
ls: cannot access '/nope': No such file or directory
```

FD Table of `ls`	
0	⌨️ keyboard
1	🖥️ display
2	🖥️ display


⌨️ keyboard
0
ls
1
🖥️ display
2
🖥️ display

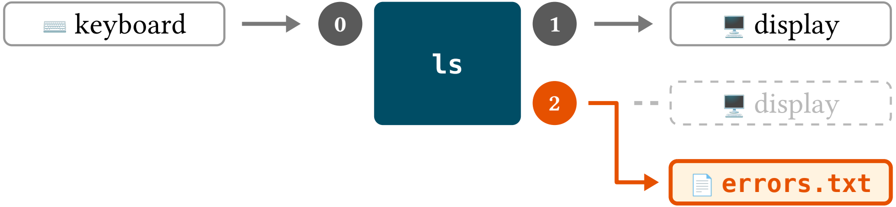
💡 errors go through `2` (`stderr`), to the display

`2>` - redirect `stderr`

The errors go into a file instead of the display

```
$ ls /nope              # the error is on the display
ls: cannot access '/nope': No such file or directory
$ ls /nope 2> errors.txt  # nothing shows up
```

FD Table of `ls`	
0	⌨️ keyboard
1	🖥️ display
2	📄 `errors.txt`



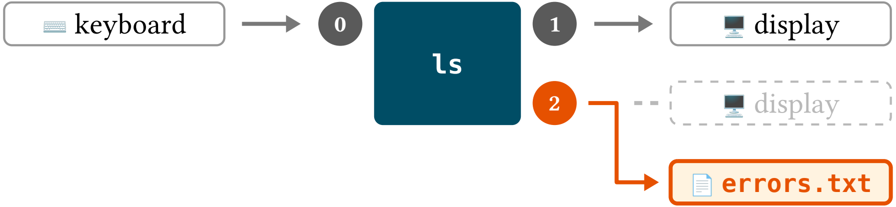
💡 `2>` **replaces** `stderr`: `2` is no longer connected to the display, it is connected to `errors.txt`

`2>` - redirect `stderr`

The errors go into a file instead of the display

```
$ ls /nope              # the error is on the display
ls: cannot access '/nope': No such file or directory
$ ls /nope 2> errors.txt  # nothing shows up
$ cat errors.txt
ls: cannot access '/nope': No such file or directory
```

FD Table of `ls`	
0	⌨️ keyboard
1	🖥️ display
2	📄 **`errors.txt`**



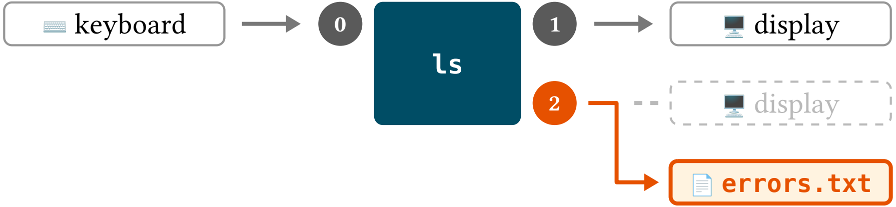
💡 `1` is not changed: `> out.txt 2> errors.txt` separates them, `&> all.txt` sends both to the same file

`2>` - redirect `stderr`

The errors go into a file instead of the display

```
$ ls /nope 2> errors.txt  # the first error
$ ls /nada 2>> errors.txt # append the error
$ cat errors.txt
ls: cannot access '/nope': No such file or directory
ls: cannot access '/nada': No such file or directory
```

FD Table of `ls`	
0	⌨️ keyboard
1	🖥️ display
2	📄 `errors.txt`



💡 `2>>` **appends**, like `>>`: the first error stays, the new one is added at the end

`|` - pipe

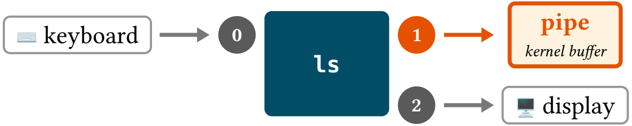
The `stdout` of a process becomes the `stdin` of the next one

pipe
kernel buffer

1. the **shell** creates a *pipe*, a buffer in the kernel with a *write* end and a *read* end

`|` - pipe

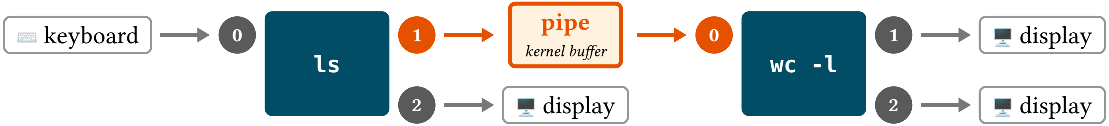
The `stdout` of a process becomes the `stdin` of the next one


⌨ keyboard
0
ls
1
pipe
kernel buffer
2
🖥 display

1. the **shell** creates a *pipe*, a buffer in the kernel with a *write* end and a *read* end
2. it starts `ls` and places the *write end* at index `1`

`|` - pipe

The `stdout` of a process becomes the `stdin` of the next one



⌨ keyboard → 0 `ls` 1 → **pipe** *kernel buffer* → 0 `wc -l` 1 → 🖥 display

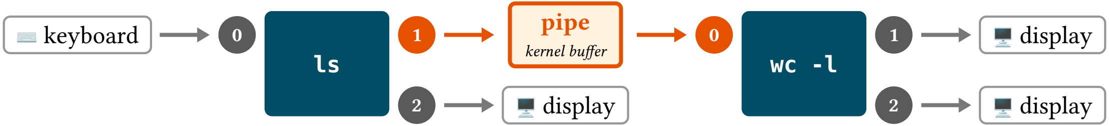
`ls` 2 → 🖥 display

`wc -l` 2 → 🖥 display

1. the **shell** creates a *pipe*, a buffer in the kernel with a *write* end and a *read* end
2. it starts `ls` and places the *write end* at index `1`
3. it starts `wc` and places the *read end* at index `0`

`|` - pipe

The `stdout` of a process becomes the `stdin` of the next one


keyboard
0
ls
1
2
pipe
kernel buffer
display
0
wc -l
1
2
display
display

1. the **shell** creates a *pipe*, a buffer in the kernel with a *write* end and a *read* end
2. it starts `ls` and places the *write end* at index `1`
3. it starts `wc` and places the *read end* at index `0`
4. both run **at the same time**: `wc` reads what `ls` writes, until `ls` ends (*end of file*)

```
$ ls
Movies  notes.txt  todo.txt
$ ls | wc -l
3
```

💡 no temporary file is used

Special Files

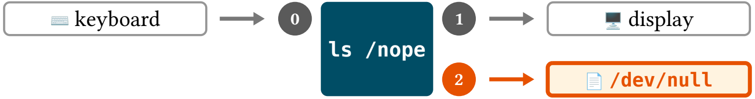
Devices made by the kernel, useful in redirects

```
/dev
├── null
├── zero
└── urandom
```

`/dev/null` - the black hole

- everything **written** is *discarded*
- **reading** gives *end of file* right away

```
$ ls /nope 2> /dev/null    # the error is discarded
$ cat < /dev/null          # nothing to read
```



Special Files

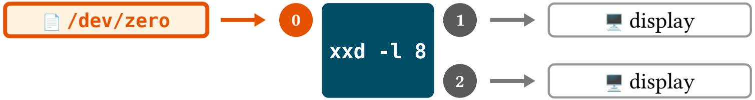
Devices made by the kernel, useful in redirects

```
/dev
├── null
├── zero
└── urandom
```

0️⃣ `/dev/zero` - endless zeros

- **reading** gives as many `0` bytes as asked
- everything **written** is *discarded*

```
$ xxd -l 8 < /dev/zero     # show the first 8 bytes
00000000: 0000 0000 0000 0000          ........
```


📄 /dev/zero
0
xxd -l 8
1
🖥️ display
2
🖥️ display

Special Files

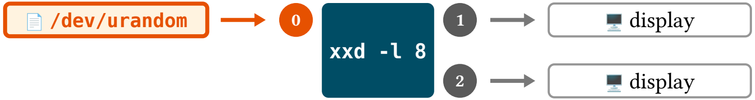
Devices made by the kernel, useful in redirects

```
/dev
├── null
├── zero
└── urandom
```

🎲 `/dev/urandom` - endless random bytes

- **reading** gives as many *random* bytes as asked
- used for keys, passwords, random file names

```
$ xxd -l 8 < /dev/urandom
00000000: 9f3a 51c2 07e8 b4d1            .:Q.....
$ xxd -l 8 < /dev/urandom # different every time
00000000: 4be0 1d77 c3a9 0f62            K..w...b
```



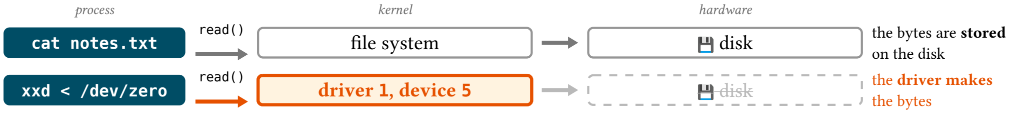
Special Files

Devices made by the kernel, useful in redirects

```
/dev
├── null
├── zero
└── urandom
```

all of them are *character devices*

- `c` instead of `-` (file) or `d` (directory), **not files**: nothing is stored on the disk
- `1, 3` - the *driver* (major `1`) and the device (minor `3`) that answer


process
kernel
hardware
cat notes.txt
read()
file system
💾 disk
xxd < /dev/zero
read()
driver 1, device 5
💾 ~~disk~~

the bytes are **stored** on the disk

the **driver makes** the bytes

```
$ ls -ld /etc/hostname /dev /dev/null /dev/zero /dev/urandom
-rw-r--r--  1 root root    7 Oct  2 09:12 /etc/hostname  ← regular file
drwxr-xr-x 18 root root 3960 Oct  2 09:12 /dev           ← directory
crw-rw-rw-  1 root root 1, 3 Oct  2 09:12 /dev/null      ← character device
crw-rw-rw-  1 root root 1, 5 Oct  2 09:12 /dev/zero      ← character device
crw-rw-rw-  1 root root 1, 9 Oct  2 09:12 /dev/urandom   ← character device
```

Redirects Cheat Sheet

Trick processes about the keyboard, display and error display

```
# Redirect stdout (standard output) to a file
ls > files.txt                      # Save the list of files into files.txt

# Append stdout to a file instead of overwriting it
ls >> files.txt                     # Add more output to files.txt

# Redirect stderr (standard error) to a file
ls /nonexistent 2> errors.txt       # Save only the error message to errors.txt

# Redirect both stdout and stderr to the same file
ls /etc /nonexistent > all.txt 2>&1 # Combine normal output and errors into all.txt

# Shortcut for redirecting both stdout and stderr (Bash 4+)
ls /etc /nonexistent &> all.txt     # Equivalent to the previous command

# Redirect stdin (standard input) from a file
sort < input.txt                    # Read lines from input.txt and sort them

# Redirect stdin and stdout at once
sort < unsorted.txt > sorted.txt    # Sort contents of unsorted.txt into sorted.txt
```

Redirects Cheat Sheet

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Redirects Cheat Sheet

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# Redirect stderr (standard error) to a file
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# Redirect stdin and stdout at once
sort < unsorted.txt > sorted.txt          # Sort contents of unsorted.txt into sorted.txt
```

Advanced Redirects

Trick processes about the keyboard, display and error display

```
# Suppress all output (stdout and stderr)
command > /dev/null 2>&1            # Run silently; discard all output and errors

# Redirect stderr to stdout and pipe to another command
command 2>&1 | grep "warning"       # Merge both streams, then search for "warning"

# Use a heredoc (redirect stdin from inline text)
cat <<EOF                           # 'cat' reads the next lines, up to EOF
Hello, world!
This is a heredoc example.
EOF
```

Advanced Redirects

Trick processes about the keyboard, display and error display

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Advanced Redirects

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cat <<EOF                               # 'cat' reads the next lines, up to EOF
Hello, world!
This is a heredoc example.
EOF
```

💡 the lines of a heredoc are sent *as they are* to `stdin`, a `#` there is not a comment

Advanced Redirects

Trick processes about the keyboard, display and error display

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command > /dev/null 2>&1            # Run silently; discard all output and errors

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```

Soft Links

Shortcuts to files and folders

Bibliography

For this section

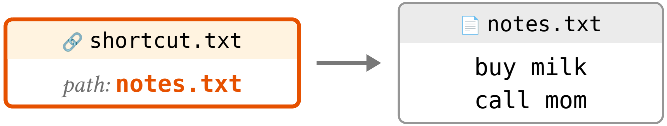
1. **Brian Ward**, *How LINUX Works*, 3rd Edition, No Starch Press, 2021
   - Chapter 2 - *Basic Commands and Directory Hierarchy*
     - Section 2.17.2 - *Working with Symbolic Links*
2. **Razvan Deaconescu, Razvan Rughinis, Mihai Carabas, Alexandru Radovici**, *Utilizarea Sistemelor de Operare*, Printech 2021, download
   - Chapter 2 - *Utilizarea sistemului de fișiere*
     - Section 2.3.4.1 - *Crearea legăturilor simbolice*

A Shortcut

A file that points to another file

```
SYNOPSIS
LN(1)   ln [OPTION]... [-T] TARGET LINK_NAME
```

```
$ ln -s notes.txt shortcut.txt    # create the link
```


🔗 shortcut.txt
path: notes.txt
📄 notes.txt
buy milk
call mom

💡 `ln -s` creates a **soft link** (*symbolic link*)

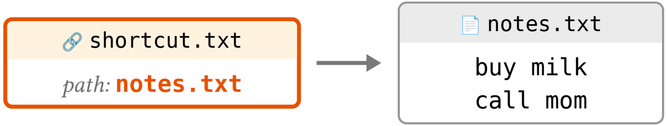
`LINK_NAME`: a **shortcut** to `TARGET`

A Shortcut

A file that points to another file

```
SYNOPSIS
LN(1)  ln [OPTION]... [-T] TARGET LINK_NAME
```

```
$ ln -s notes.txt shortcut.txt   # create the link
$ ls -l
-rw-r--r-- 1 alice alice 18 Oct 12 09:12 notes.txt
lrwxrwxrwx 1 alice alice  9 Oct 12 09:12 shortcut.txt -> notes.txt
```


🔗 shortcut.txt
path: notes.txt
📄 notes.txt
buy milk
call mom

💡 `l` - the type is *link*, and `ls -l` shows where it points: `-> notes.txt`

A Shortcut

A file that points to another file

```
SYNOPSIS
LN(1)  ln [OPTION]... [-T] TARGET LINK_NAME
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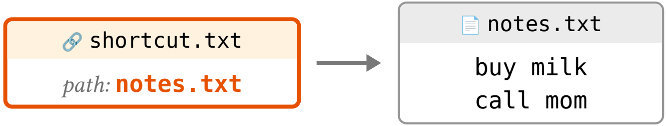
```
$ ln -s notes.txt shortcut.txt   # create the link
$ ls -l
-rw-r--r-- 1 alice alice 18 Oct 12 09:12 notes.txt
lrwxrwxrwx 1 alice alice  9 Oct 12 09:12 shortcut.txt -> notes.txt
$ cat shortcut.txt               # reads notes.txt
buy milk
call mom
```


🔗 shortcut.txt
path: notes.txt
📄 notes.txt
buy milk
call mom

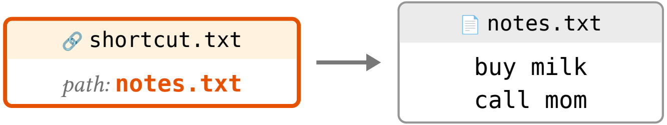
💡 using the link uses the **target**: `cat` reads the lines of `notes.txt`

The Contents of a Link

Just a path to a file or a folder

```
$ readlink shortcut.txt
notes.txt
```

- the link does **not** hold the data, it holds a **path**
- the size of `shortcut.txt` is `9`: the length of the text `notes.txt`



🔗 shortcut.txt

path: **notes.txt**

→

📄 notes.txt

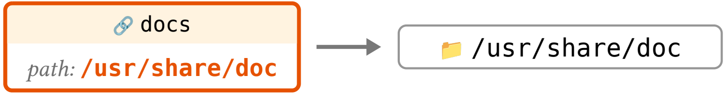
```
buy milk
call mom
```

The Contents of a Link

Just a path to a file or a folder

```
$ readlink shortcut.txt
notes.txt
$ ln -s /usr/share/doc docs
$ readlink docs
/usr/share/doc
```

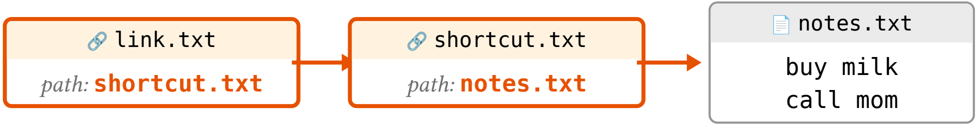
- the link does **not** hold the data, it holds a **path**
- the size of `shortcut.txt` is `9`: the length of the text `notes.txt`
- the target can be a **file** or a **folder**
- the path can be *absolute* (`/usr/share/doc`) or *relative* to the folder of the link (`notes.txt`)



Following Links

The kernel reads the path and opens the target instead

```
$ ln -s shortcut.txt link.txt    # a link to a link
$ cat link.txt
buy milk
call mom
```



🔗 link.txt

path: **shortcut.txt**

🔗 shortcut.txt

path: **notes.txt**

📄 notes.txt

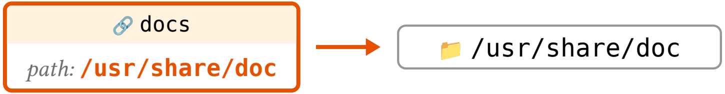
buy milk
call mom

💡 a link can point to **another link**: the kernel follows them all, until it finds a real file

Following Links

The kernel reads the path and opens the target instead

```
$ ln -s shortcut.txt link.txt    # a link to a link
$ cat link.txt
buy milk
call mom
$ ls docs/                       # into the folder
bash  coreutils  git  ...
```

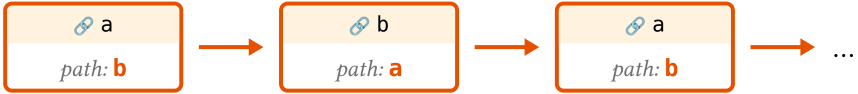

🔗 docs
path: /usr/share/doc
📁 /usr/share/doc

💡 a link to a folder works like the folder: `ls`, `cd`

Following Links

Loops, and commands that do not follow

```
$ ln -s a b                         # b -> a
$ ln -s b a                         # a -> b, a loop
$ cat a
cat: a: Too many levels of symbolic links
```

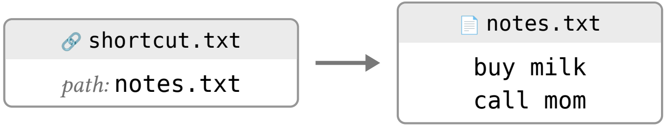


💡 a loop never gets to a real file: the kernel gives up after a few links

Following Links

Loops, and commands that do not follow

```
$ ln -s a b                   # b -> a
$ ln -s b a                   # a -> b, a loop
$ cat a
cat: a: Too many levels of symbolic links
$ rm link.txt                 # deletes only the link
$ cat shortcut.txt
buy milk
```



🔗 shortcut.txt

path: `notes.txt`

📄 notes.txt

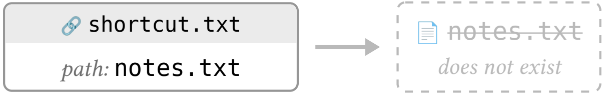
buy milk
call mom

⚠️ `ls -l`, `readlink` and `rm` do **not** follow: they work on the **link itself**

Broken Links

The target is gone, the link stays

```
$ rm notes.txt                  # delete the target
```



🔗 `shortcut.txt`

path: `notes.txt`

📄 ~~`notes.txt`~~

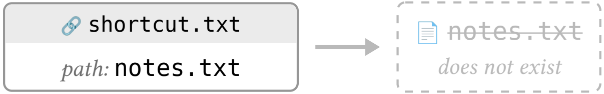
does not exist

💡 deleting the target does **not** delete the links to it

Broken Links

The target is gone, the link stays

```
$ rm notes.txt                        # delete the target
$ ls -l shortcut.txt
lrwxrwxrwx 1 alice alice  9 Oct 12 09:12 shortcut.txt -> notes.txt
```

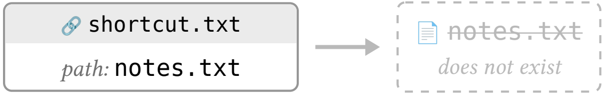

🔗 shortcut.txt
path: notes.txt
📄 ~~notes.txt~~
does not exist

💡 the link still holds its path, but `notes.txt` does not exist: the link is **broken**

Broken Links

The target is gone, the link stays

```
$ rm notes.txt                    # delete the target
$ ls -l shortcut.txt
lrwxrwxrwx 1 alice alice  9 Oct 12 09:12 shortcut.txt -> notes.txt
$ cat shortcut.txt
cat: shortcut.txt: No such file or directory
```


🔗 shortcut.txt
path: notes.txt
📄 ~~notes.txt~~
does not exist

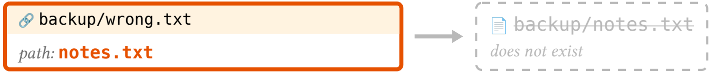
- 💡 following a broken link fails
- ⚠️ `ln -s` does not check that the target exists

Relative Paths

They start from the folder **of the link**, not from the folder you are in

```
$ cd /home/alice
$ ln -s notes.txt backup/wrong.txt      # from backup/
$ cat backup/wrong.txt
cat: backup/wrong.txt: No such file or directory
```

```
/home/alice
├── notes.txt
└── backup
    └── wrong.txt
```


🔗 backup/wrong.txt
path: **notes.txt**
📄 ~~backup/notes.txt~~
does not exist

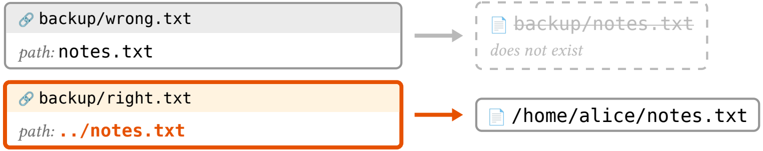
💡 `wrong.txt` is in `backup/`, so the kernel looks for `notes.txt` **in `backup/`**: the link is broken from the start

Relative Paths

They start from the folder **of the link**, not from the folder you are in

```
$ cd /home/alice
$ ln -s notes.txt backup/wrong.txt     # from backup/
$ cat backup/wrong.txt
cat: backup/wrong.txt: No such file or directory
$ ln -s ../notes.txt backup/right.txt  # go up
$ cat backup/right.txt
buy milk
```

```
/home/alice
├── notes.txt
└── backup
    ├── wrong.txt
    └── right.txt
```



🔗 `backup/wrong.txt`

path: `notes.txt`

📄 ~~`backup/notes.txt`~~

does not exist

🔗 `backup/right.txt`

path: **`../notes.txt`**

📄 `/home/alice/notes.txt`

💡 `../notes.txt` goes up from `backup/` to `/home/alice`: the file is found

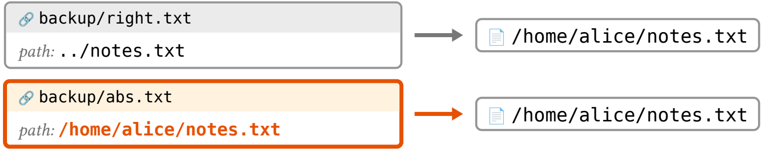
⚠️ write a relative path as if you were in the **folder of the link**

Moving a Link

The path stays the same, the folder it starts from changes

```
$ ln -s /home/alice/notes.txt backup/abs.txt
```

```
/home/alice
├── notes.txt
└── backup
    ├── right.txt
    └── abs.txt
```



🔗 backup/right.txt

path: `../notes.txt`

→ 📄 /home/alice/notes.txt

🔗 backup/abs.txt

path: **`/home/alice/notes.txt`**

→ 📄 /home/alice/notes.txt

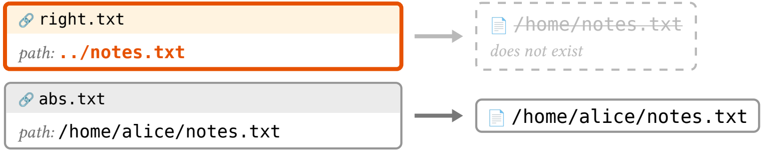
💡 two links in `backup/` to the same file: one with a *relative* path, one with an *absolute* path

Moving a Link

The path stays the same, the folder it starts from changes

```
$ ln -s /home/alice/notes.txt backup/abs.txt
$ mv backup/right.txt backup/abs.txt .
$ cat right.txt
cat: right.txt: No such file or directory
```

```
/home/alice
├── notes.txt
├── right.txt  <- moved here
├── abs.txt  <- moved here
└── backup
~~    ├── right.txt~~
~~    └── abs.txt~~
```



🔗 right.txt

path: **`../notes.txt`**

📄 ~~/home/notes.txt~~

does not exist

🔗 abs.txt

path: `/home/alice/notes.txt`

📄 /home/alice/notes.txt

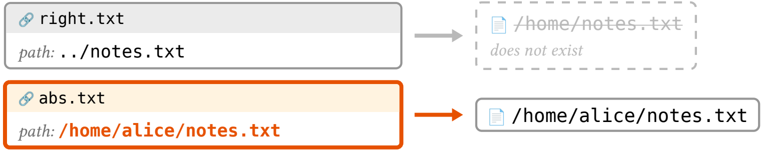
💡 `mv` moves both links up; `right.txt` still holds `../notes.txt`, but now it starts from `/home/alice`: the link is **broken**

Moving a Link

The path stays the same, the folder it starts from changes

```
$ ln -s /home/alice/notes.txt backup/abs.txt
$ mv backup/right.txt backup/abs.txt .
$ cat right.txt
cat: right.txt: No such file or directory
$ cat abs.txt
buy milk
call mom
```

```
/home/alice
├── notes.txt
├── right.txt  <- moved here
├── abs.txt  <- moved here
└── backup
~~    ├── right.txt~~
~~    └── abs.txt~~
```



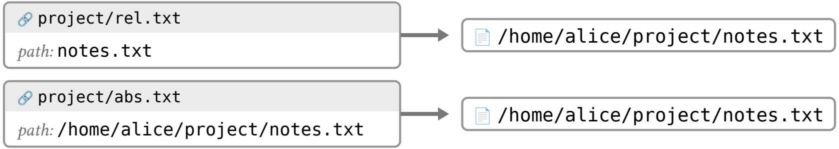
🔗 right.txt
path: ../notes.txt
📄 ~~/home/notes.txt~~
does not exist
🔗 abs.txt
path: **/home/alice/notes.txt**
📄 /home/alice/notes.txt

💡 an *absolute* path starts from `/`: it does not matter where the link is

Moving the Folder

The link and its target move together

```
/
├── home/alice
│   └── project
│       ├── notes.txt
│       ├── rel.txt
│       └── abs.txt
└── tmp
```



🔗 `project/rel.txt`

path: `notes.txt`

📄 `/home/alice/project/notes.txt`

🔗 `project/abs.txt`

path: `/home/alice/project/notes.txt`

📄 `/home/alice/project/notes.txt`

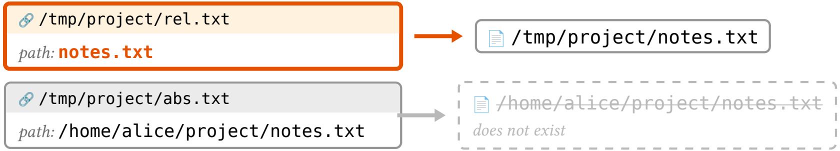
💡 `project/` holds a file and two links to it

Moving the Folder

The link and its target move together

```
$ mv project /tmp/project      # the whole folder
$ cat /tmp/project/rel.txt
buy milk
call mom
```

```
/
├── home/alice
│   └── project
└── tmp
    └── project  <- moved here
        ├── notes.txt
        ├── rel.txt
        └── abs.txt
```



🔗 `/tmp/project/rel.txt`

path: **`notes.txt`**

📄 `/tmp/project/notes.txt`

🔗 `/tmp/project/abs.txt`

path: `/home/alice/project/notes.txt`

📄 ~~`/home/alice/project/notes.txt`~~

does not exist

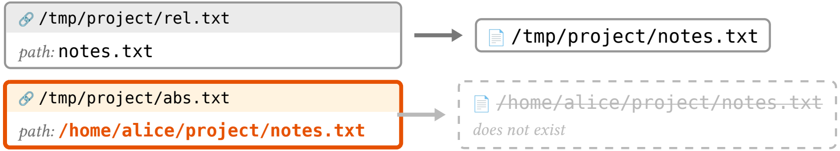
💡 `rel.txt` starts from its own folder, which moved together with `notes.txt`: it still works

Moving the Folder

The link and its target move together

```
$ mv project /tmp/project     # the whole folder
$ cat /tmp/project/rel.txt
buy milk
call mom
$ cat /tmp/project/abs.txt
cat: /tmp/project/abs.txt: No such file or directory
```

```
/
├── home/alice
~~│   └── project~~
└── tmp
    └── project  <- moved here
        ├── notes.txt
        ├── rel.txt
        └── abs.txt
```



🔗 /tmp/project/rel.txt

path: `notes.txt`

📄 /tmp/project/notes.txt

🔗 /tmp/project/abs.txt

path: **`/home/alice/project/notes.txt`**

📄 ~~/home/alice/project/notes.txt~~

does not exist

💡 `abs.txt` still points to the old place of the file: the link is **broken**

Relative or Absolute?

Choose the path by what will move

	relative path	**absolute path**
the path starts from	the folder of the link	`/`
only the **link** is moved	**broken**	works
the link and the target are moved **together**	works	**broken**
the **target** is moved or deleted	**broken**	**broken**

💡 *relative* links for files that travel together (a project, an archive), *absolute* links for files that stay in place (`/usr/share/doc`, `/etc/...`)

The `/proc` Folder

Look inside the processes

Bibliography

For this section

1. **Brian Ward**, *How LINUX Works*, 3rd Edition, No Starch Press, 2021
   - Chapter 4 - *Disks and Filesystems*
     - Section 4.2.12 - *Special-Purpose Filesystems*
   - Chapter 8 - *A Closer Look at Processes and Resource Utilization*
     - Section 8.1 - *Tracking Processes*
     - Section 8.2 - *Finding Open Files with lsof*
2. **Razvan Deaconescu, Razvan Rughinis, Mihai Carabas, Alexandru Radovici**, *Utilizarea Sistemelor de Operare*, Printech 2021, download
   - Chapter 4 - *Procese*
     - Section 4.2.2 - *Utilitare pentru urmărirea proceselor*
     - Section 4.8 - *Anexa: Sistemul de fișiere procfs*

`/proc` - the process file system

Files made by the kernel, when they are read

```
/proc
├── 1          # systemd
├── 2          # kthreadd
├── ...
├── 4100       # bash
├── 4321       # ...
├── self
├── cpuinfo
├── meminfo
├── uptime
└── ...
```

- `/proc` is a *virtual* file system: every process has a folder, its name is the **PID**

`/proc` - the process file system

Files made by the kernel, when they are read

```
/proc
├── 1           # systemd
├── 2           # kthreadd
├── ...
├── 4100        # bash
├── 4321        # ...
├── self
├── cpuinfo
├── meminfo
├── uptime
└── ...
```

```
$ ls -l /proc/uptime    # the size is 0
-r--r--r-- 1 root root 0 Oct 12 09:12 /proc/uptime
$ cat /proc/uptime      # but it has contents
244699.78 368114.32
```

- `/proc` is a *virtual* file system: every process has a folder, its name is the **PID**
- **nothing** is stored on the disk: the **kernel** writes the contents when a process **reads** the file

`/proc` - the process file system

Files made by the kernel, when they are read

```
/proc
├── 1          # systemd
├── 2          # kthreadd
├── ...
├── 4100       # bash
├── 4321       # ...
├── self
├── cpuinfo
├── meminfo
├── uptime
└── ...
```

```
$ ls -l /proc/uptime    # the size is 0
-r--r--r-- 1 root root 0 Oct 12 09:12 /proc/uptime
$ cat /proc/uptime      # but it has contents
244699.78 368114.32
$ readlink /proc/self   # the PID of readlink
4325
```

- `/proc` is a *virtual* file system: every process has a folder, its name is the **PID**
- **nothing** is stored on the disk: the **kernel** writes the contents when a process **reads** the file
- `/proc/self` - the folder of the process that reads it: `ls /proc/self` lists the folder of `ls`

`/proc/<PID>` - inside a process

The resources of a process, as files

```
/proc/4321
├── cmdline   # the arguments
├── comm      # the name
├── cwd       # -> the current folder
├── environ   # the environment variables
├── exe       # -> the executable file
├── fd/       # the File Descriptor Table
├── maps      # the memory map
└── status    # PID, PPID, state, ...
```

🪪 Identification

- `status` has the *PID*, the *PPID* and the *state* of the process

`/proc/<PID>` - inside a process

The resources of a process, as files

```
/proc/4321
├── cmdline   # the arguments
├── comm      # the name
├── cwd       # -> the current folder
├── environ   # the environment variables
├── exe       # -> the executable file
├── fd/       # the File Descriptor Table
├── maps      # the memory map
└── status    # PID, PPID, state, ...
```

📄 The program

- `exe` - a link to the executable that was loaded (`exec`)
- `cmdline` - the arguments it got
- `comm` - its name

`/proc/<PID>` - inside a process

The resources of a process, as files

```
/proc/4321
├── cmdline   # the arguments
├── comm      # the name
├── cwd       # -> the current folder
├── environ   # the environment variables
├── exe       # -> the executable file
├── fd/       # the File Descriptor Table
├── maps      # the memory map
└── status    # PID, PPID, state, ...
```

💾 Data

- `fd/` - one link for every open file descriptor
- `cwd` - a link to the current folder

`/proc/<PID>` - inside a process

The resources of a process, as files

```
/proc/4321
├── cmdline   # the arguments
├── comm      # the name
├── cwd       # -> the current folder
├── environ   # the environment variables
├── exe       # -> the executable file
├── fd/       # the File Descriptor Table
├── maps      # the memory map
└── status    # PID, PPID, state, ...
```

🧠 Memory

- `maps` - where the code, the libraries and the variables are in memory
- `environ` - the environment variables

`/proc/<PID>` - inside a process

The resources of a process, as files

```
/proc/4321
├── cmdline   # the arguments
├── comm      # the name
├── cwd       # -> the current folder
├── environ   # the environment variables
├── exe       # -> the executable file
├── fd/       # the File Descriptor Table
├── maps      # the memory map
└── status    # PID, PPID, state, ...
```

```
$ readlink /proc/self/exe
/usr/bin/readlink
$ cd /tmp
$ readlink /proc/self/cwd
/tmp
```

`/proc/<PID>/status`

What the kernel knows about a process

```
$ cat /proc/self/status
Name:     cat
Umask:    0022
State:    R (running)
...
Pid:      4330
PPid:     4100
...
Threads:  1
...
```

- `State` - one of the *process states*
- `Threads` - how many threads run in parallel

`/proc/<PID>/status`

What the kernel knows about a process

```
$ cat /proc/self/status
Name:     cat
Umask:    0022
State:    R (running)
...
Pid:      4330
PPid:     4100
...
Threads:  1
...
$ echo $$     # the PID of the shell
4100
```

- `State` - one of the *process states*
- `Threads` - how many threads run in parallel
- the `PPid` of `cat` is the shell: the shell called `fork` and `exec` to start it

`/proc/<PID>/status`

What the kernel knows about a process

```
$ cat /proc/self/status
Name:     cat
Umask:    0022
State:    R (running)
...
Pid:      4330
PPid:     4100
...
Threads:  1
...
$ echo $$     # the PID of the shell
4100
```

- `State` - one of the *process states*
- `Threads` - how many threads run in parallel
- the `PPid` of `cat` is the shell: the shell called `fork` and `exec` to start it

💡 `ps`, `top` and `pstree` do not ask the kernel directly: they **read the files** in `/proc`

`/proc/<PID>/fd` - the File Descriptor Table

Every open file descriptor is a link

```
$ ls -l /proc/self/fd
total 0
lrwx------ 1 alice alice 64 Oct 12 09:12 0 -> /dev/pts/0
lrwx------ 1 alice alice 64 Oct 12 09:12 1 -> /dev/pts/0
lrwx------ 1 alice alice 64 Oct 12 09:12 2 -> /dev/pts/0
```

FD Table of `ls`	
⓪	📄 `/dev/pts/0`
①	📄 `/dev/pts/0`
②	📄 `/dev/pts/0`

💡 `0`, `1` and `2` point to `/dev/pts/0`, the *terminal*: it is both the keyboard and the display

`/proc/<PID>/fd` - the File Descriptor Table

Every open file descriptor is a link

```
$ ls -l /proc/self/fd
total 0
lrwx------ 1 alice alice 64 Oct 12 09:12 0 -> /dev/pts/0
lrwx------ 1 alice alice 64 Oct 12 09:12 1 -> /dev/pts/0
lrwx------ 1 alice alice 64 Oct 12 09:12 2 -> /dev/pts/0
lr-x------ 1 alice alice 64 Oct 12 09:12 3 -> /proc/4331/fd
```

FD Table of `ls`	
0	📄 `/dev/pts/0`
1	📄 `/dev/pts/0`
2	📄 `/dev/pts/0`
3	📄 `/proc/4331/fd`

💡 `ls` opened the folder it lists, at the first free index: `3`, just like `sort` opened `fruits.txt`

`/proc/<PID>/fd` - the File Descriptor Table

Every open file descriptor is a link

```
$ lsof -p 4100      # the open files of the shell (PID 4100)
COMMAND  PID USER   FD  TYPE DEVICE SIZE/OFF NODE NAME
...
bash    4100 alice  0u  CHR  136,0      0t0    3 /dev/pts/0
bash    4100 alice  1u  CHR  136,0      0t0    3 /dev/pts/0
bash    4100 alice  2u  CHR  136,0      0t0    3 /dev/pts/0
```

💡 `lsof -p <PID>` lists the same information: it reads `/proc/<PID>/fd`

⚠️ only the **owner** of the process (or `root`) can look inside its `fd/`

💡 a redirect changes these links: with `> out.txt`, `1` points to `out.txt`

`/proc` - the whole system

Not only processes

```
/proc
├── cpuinfo    # the processors
├── meminfo    # the memory
├── uptime     # seconds since boot
├── loadavg    # how busy the system is
└── mounts     # the mounted file systems
```

file	shown by
`/proc/cpuinfo`	`lscpu`
`/proc/meminfo`	`free`
`/proc/uptime`, `/proc/loadavg`	`uptime`
`/proc/mounts`	`mount`

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$ head -3 /proc/meminfo
MemTotal:       65296068 kB
MemFree:        10651832 kB
MemAvailable:   53764732 kB
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💡 most of the system tools just **read** `/proc` and format the result nicely

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We talked about

- Installing applications
- Process
- PID
- Process Loading
- Process States
- `fork` and `exec`
- How Linux starts
- Orphan and zombie processes
- File descriptors
- Redirects
- Soft links
- The `/proc` folder